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# Block rotations and continental extension in the central Aegean Sea: palaeomagnetic and structural evidence from Tinos and Mykonos (Cyclades, Greece)

Dov Avigad [a,*], Gidon Baer [b], Ariel Heimann [b]

[a] *Institute of Earth Sciences, The Hebrew University, Jerusalem 91904, Israel*
[b] *Israel Geological Survey, 30 Malchei Israel St., Jerusalem, Israel*

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**Abstract**

The Aegean Sea, floored by an attenuated continental crust, is a Mediterranean-type back-arc basin formed above the subducting African plate. We investigated post-12 Ma block rotation in relation to back-arc extension in the area of the western Cyclades (central Aegean Sea). On Tinos island, NW-trending dacitic dikes and a Miocene monzogranite penetrated an Alpine metamorphic series. K–Ar analyses of 6 dike samples yielded an average age of 11.5 ± 0.4 Ma. The palaeomagnetic analyses of the dikes and the monzogranite reveal 23° vertical-axis clockwise rotation. We assume that the dacitic dikes intruded vertical (as they are today) and that no tilt correction is required. This is supported by the fact that no inclination anomaly is revealed when the palaeomagnetic data is compared with a tilt-corrected pole position from Kimi (Evia). The absence of tilting implies that post-12 Ma crustal thinning in the area of Tinos island was minor. On Mykonos, a low-angle normal fault dipping 30° to the NE juxtaposes a series of Oligo-Miocene molasse sediments above a 10–12-Ma granite. Published palaeomagnetic data revealed a considerable amount of inclination flattening and 22° clockwise rotation of the footwall granite. We show that tilting predated vertical-axis rotation, and that the (now) low-angle normal fault originated steeper, dipping ca. 54°. Planar-rotational fault geometry indicates extension of ca. 60%. In spite of the significant variations in the magnitude of post-12 Ma upper crustal extension across the Tinos–Mykonos transect, crustal thickness below the entire area changes only slightly. This indicates that upper-crustal levels were decoupled from the lower crust. The widespread Middle Miocene granitic magmatism in the Cyclades suggests that flow in the lower crust may have had a dominant role in maintaining the smooth Moho. Assuming that the western Cyclades constitute a set of NW-striking crustal blocks, it is estimated that vertical-axis rotations increased the N–S dimension of the area by ca. 31% (e.g. 25 km). The exact timing of this clockwise rotation is not well constrained, but it postdated the formation of low-angle normal faults and tilting, and may have taken place in the Late Miocene. © 1998 Elsevier Science B.V. All rights reserved.

*Keywords:* extension; paleomagnetism; geochronology; tectonics; rotation; back-arc basins; Aegean Sea; Cyclades; Greece

* Corresponding author. On leave at Laboratoire de Géologie, ENS, 24 rue Lhomond 75005 Paris, France. E-mail: avigad@euclase.ens.fr

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## 1. Introduction

The Aegean Sea was formed on the site of a previous alpine orogenic belt that existed in the area at the beginning of the Tertiary. The Aegean Sea is floored by an attenuated continental crust (28–30 km thick below the Cyclades [1]) and is currently being extended in a N–S direction at a rate of ~4 cm/y ([2] and references therein). In the central Aegean, in the area of the Cyclades (Fig. 1), the reworking of the Alpine orogenic belt by extensional tectonics resulted in widespread exposures of eclogite-facies rocks in a terrane dominated by low- and high-angle normal faults [3–6]. Most of the rocks exposed in the Cycladic islands (Fig. 1) are Alpine metamorphic rocks and Miocene granites [7,8]. Miocene sediments are confined to tiny allochthonous sequences above low-angle normal faults whose footwalls are composed of metamorphic rocks and Miocene granitoids [3,4].

Extension in the Aegean is being promoted by the southward retreat of the Hellenic Trench [9], and is thought to accommodate right-lateral motion on the North Anatolian Fault that invades the northern Aegean region in relation to the overall westward extrusion of Turkey [10]. It was estimated that back-arc extension in the Aegean region commenced at around 13 Ma [9], roughly coeval with the onset of right-lateral motion on the North Anatolian Fault. Yet, more recent studies on metamorphic core-complexes exposed in the Cyclades suggested that back-arc extension began at least in the Early

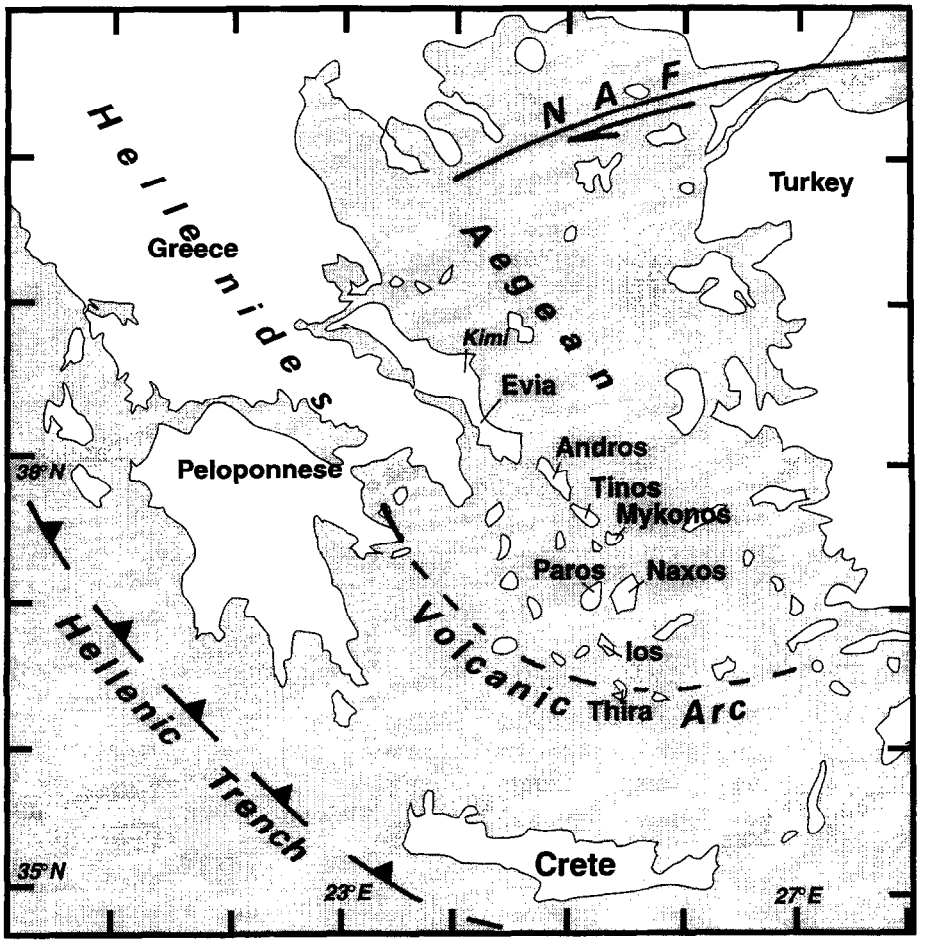



Fig. 1. Location map of the Aegean region showing the area of the Cycladic islands and sites mentioned in the text.

Miocene, at around 20–25 Ma [3] in association with a regional scale greenschist-facies retrograde overprint of the Cycladic blueschists at 20–25 Ma [8,11,12]. Structures associated with this overprint were interpreted to reflect crustal-scale extension [3,5,13]. Kinematic criteria in the metamorphic rocks and in the sheared Miocene granitoids generally show top-to-the-ENE, or to-the-NE sense of motion ([5,14] and references therein).

Although the style and kinematics of the extensional structures associated with the greenschist-facies overprint at around 20–25 Ma are relatively well documented, very little is, in fact, known about the properties of the extensional deformation that affected the Cyclades later in the Miocene. Low-angle normal faults that cut through Middle Miocene granites are abundant in the central part of the Cyclades, but the origin of these low-angle normal faults, the amount of extension involved, the scale of block tilting and rotations, and the spatial and temporal distribution of the extensional strain are still poorly understood.

Important clues on the post-Middle Miocene tectonic evolution of the Aegean region were derived from palaeomagnetic studies which were, until recently, confined to the periphery of the Aegean Sea. These studies revealed Neogene clockwise rotations on vertical axes, of up to 50°, in the western side of the Aegean ([15,16] and references therein), whereas on the Turkish side, both clockwise and anticlockwise tectonic rotations were detected. Zero rotations were detected on Crete [16]. On the basis of this rotation pattern, Kissel and Laj [16] depicted the opening history of the Aegean Sea and the development of the Hellenic arc since the Middle Miocene. Yet, data on block rotations in the central part of the Aegean Sea, in the area of the Cyclades that bridges Mainland Greece with Turkey, was not available until a recent study by Morris and Anderson [17] revealed opposite-sense, vertical-axis rotations of significant magnitudes on Naxos and Mykonos (Fig. 2). Morris and Anderson suggested that Mykonos belongs to a NW-striking block that rotated 22° clockwise, while Naxos, which shows 30° anticlockwise rotation, is tied to a NE-striking crustal segment perhaps connected to the Turkish side of the Aegean Sea.

Our study on Tinos island reveals more evidence on the role of rigid block rotations in shaping the post-Middle Miocene structure of the central Aegean region. We focus on a unique swarm of NW-trending, sub-vertical dikes that penetrated the metamorphic rocks exposed on this island. These dikes, striking perpendicular to the overall stretching direction, are a useful planar marker that can monitor the mode of rigid block rotations in the area. We also integrate field observations with previously published palaeomagnetic data obtained on a 12-Ma granite on Mykonos [16], in order to estimate the mode of crustal deformation and the origin of Middle Miocene low-angle normal fault exposed there. The combined results from Tinos and Mykonos provide a better picture of how post-Middle Miocene strain was partitioned in the central Aegean and what were the dominant modes of deformation in each of these islands.

## 2. Tinos island

### 2.1. Geological and structural setting

The island of Tinos (Fig. 2A), 200 km$^2$, is the third largest island of the Cyclades and forms a 30-km NW–SE-striking elongated ridge. The rocks exposed on Tinos [12,18,19] comprise Alpine metamorphic units, an Early Miocene monzogranite, a swarm of Miocene dacitic dikes, and tiny outcrops of Mio–Pliocene clastic sediments on the NE flank of the island. The SW coast of the island is a steep cliff rising to a height of ca. 600 m above sea level. The top surface of the island is gently dipping to the NE and gradually plunges towards the sea. The overall morphology suggests that the island is slightly tilted to the northeast and that it is bounded by steep NW–SE-trending faults parallel to its morphological axis.

The Alpine metamorphic rocks on Tinos include the Cycladic blueschist unit metamorphosed at high-pressure conditions in the Eocene and subsequently overprinted in the greenschist facies at around 20 Ma [12], and an overlying low-pressure unit comprising dismembered ophiolites and phyllites [20] yielding Upper Cretaceous K–Ar ages [21]. The low-angle tectonic contact between the Upper Unit and the blueschist unit (Fig. 2B) juxtaposed low-pressure metamorphic rocks over blueschists and

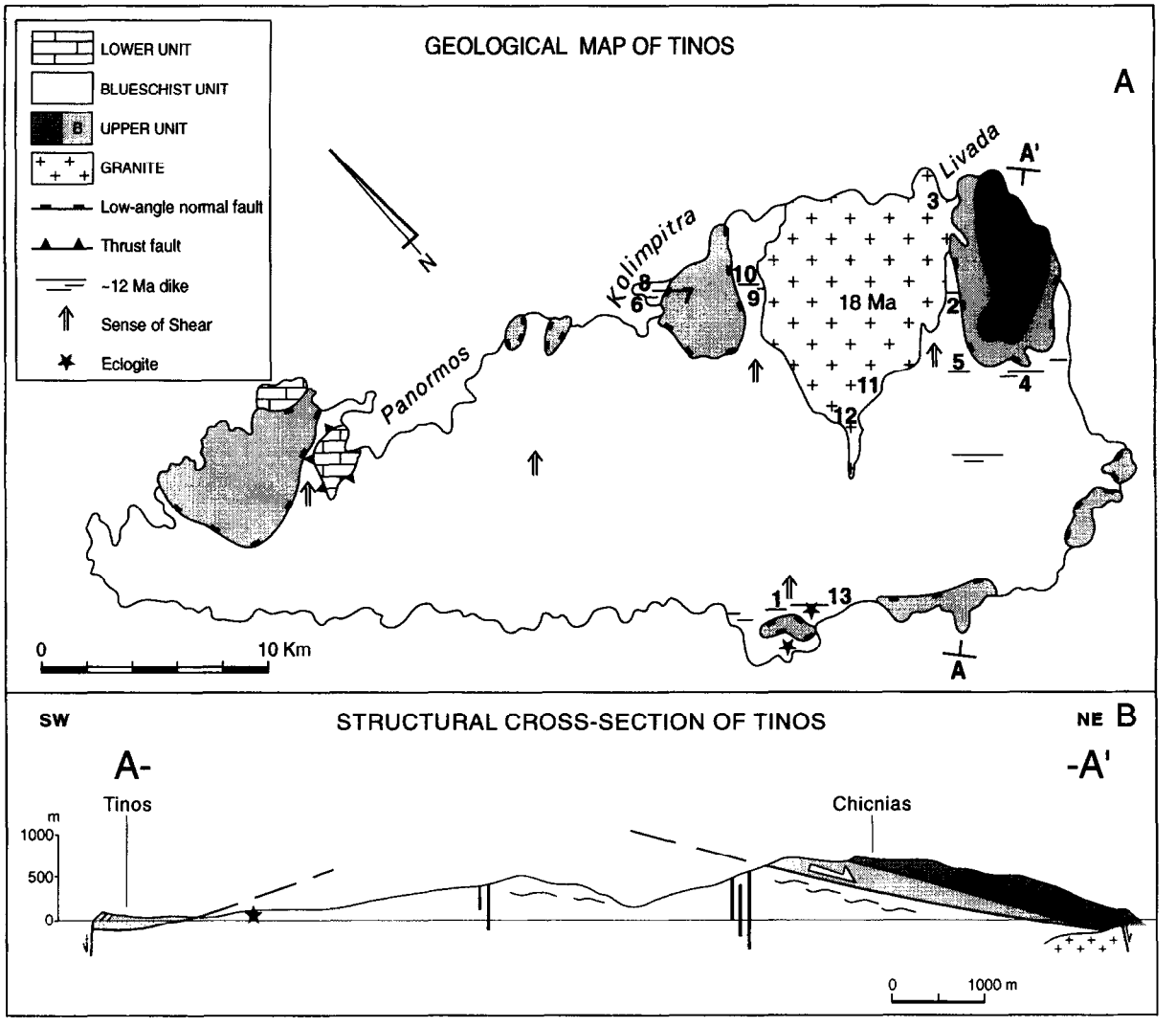



Fig. 2. (A) Geological map of Tinos island [18,19]. Numbers *1–13* correspond to palaeomagnetic and geochronological sites on Tables 1 and 2. Note the presence of dikes in both Blueschist and Upper units. Arrows mark sense of ductile shear in the footwall ([12], and our own observations). (B) A structural cross section (*A–A'* in A) of the island of Tinos shows the main detachment surface that separates the LP Upper unit in the hangingwall from the underlying Cycladic blueschist unit. This is the oldest known extensional discontinuity in the Cyclades. Note that the Early Miocene granite pierced the detachment in the area of Livada.

eclogites and was interpreted as an extensional detachment [19]. The sub-horizontal layering in the hangingwall and the absence of tilting in it suggest that the low-angle normal fault on Tinos originated shallow dipping. Semi-ductile shear zones and 'C–S' fabrics in the Blueschist unit at the footwall, indicate a prolonged extensional history with a top-to-the-NE sense of motion ([5,14], and our own data).

A large monzogranitic pluton intruded the metamorphic sequence (Fig. 2A,B) piercing the extensional detachment [19]. K–Ar ages of ca. 15–18 Ma were obtained [8] indicating that this monzogranite is the oldest among the group of Miocene granites that intruded the central Aegean. The granite is exposed in a NE-striking antiform and is weakly deformed showing non-penetrative NE lineation in its eastern part.

A swarm of NW–SE-trending dikes was traced by us in the southern half of Tinos island (Fig. 2A). The dikes are mostly vertical (Fig. 2B) or weakly inclined (up to 5°) towards SW or NE. They are up to several meters thick, and can be traced over hundreds of meters along strike (Fig. 2A). The dikes cut the nearly flat cleavage and shear fabric of the metamorphic rocks and, in several places, they also penetrate the detachment and cut through a serpentinite-bearing sequences of the hangingwall Upper Unit (near Kolimpitra Bay, Fig. 2A).

The dikes consist of variable amounts of hornblende and plagioclase phenocrysts with a lesser amount of biotite with feldspar and quartz in the matrix. Whole-rock chemical analyses of five dike samples are presented on a Total Alkalis–Silica plot (Fig. 3). Most dikes have dacitic composition. They are probably remnants of a larger magmatic edifice formed in relation to the opening of the Aegean back-arc [22].

In the following sections we present K–Ar ages and the palaeomagnetic vectors of these dikes and of the monzodiorite pluton exposed in Tinos.

### 2.2. Geochronology

K–Ar age determinations were made in the geochronological laboratory of the Geological Survey of Israel (Jerusalem). Six samples of dacitic dikes from the island of Tinos were measured (sites on Fig. 2A). Analytical techniques were as in Heimann and Ron [23].

Our K–Ar dating of these dikes (Table 1) yielded a mean age of 11.5 ± 0.43 Ma indicating that this dike swarm witnessed a substantial part of the Miocene-to-Recent extensional deformation in the central Aegean region, and that they are an excellent passive planar marker to assess the amount of post-12 Ma horizontal- and vertical-axis rotations using palaeomagnetism.

One granite sample from Tinos yielded an age of 15.9 ± 0.35 Ma for amphibole concentrate, and 14.98 ± 0.31 Ma for biotite concentrate. These ages are consistent with previous data by Altherr et al. [8], who also provide fission track ages on sphene and apatite of 13.8 and 10.8 Ma, interpreted as cooling below 280 and 120°C, respectively.

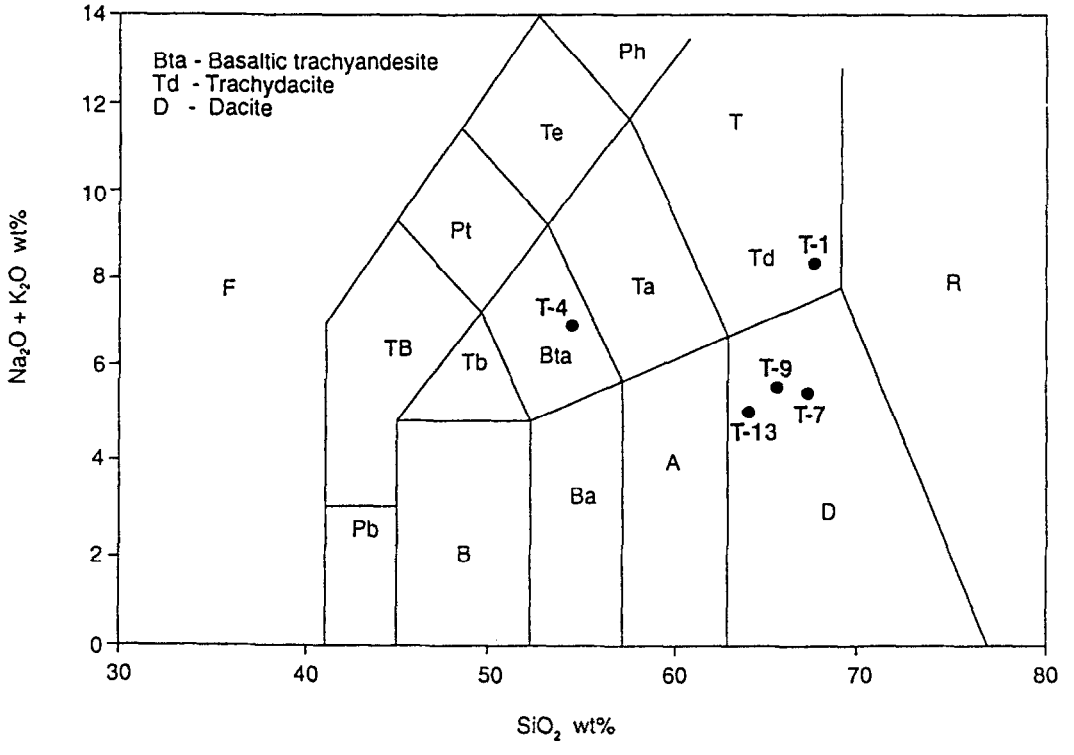



Fig. 3. Total alkalis versus silica plot for the Tinos dikes. Sample numbers corresponds to sites on Fig. 2a.

Table 1
K–Ar analytical data for dacitic dikes and granite from Tinos island

| Site no. | WR or MC [a] | %K | $\%^{40}Ar^{*}$ [b] | $^{40}Ar^{*}$ [c] | Age (Ma) | Error (Ma) | Mean age (Ma) |
|---|---|---|---|---|---|---|---|
| *Dacitic dikes* | | | | | | | |
| T1 | WR | 1.33 | 76.13 | 0.587 | 11.32 | 0.24 | |
| T2 | WR | 4.6 | 64.69 | 21.89 | 12.20 | 0.30 | |
| | WR | 4.6 | 84.24 | 2.208 | 12.31 | 0.25 | |
| T4 | WR | 4.5 | 82.73 | 2.009 | 11.45 | 0.25 | |
| T7 | WR | 6.55 | 46.83 | 2.849 | 11.16 | 0.23 | |
| T9 | BC | 3.40 | 54.51 | 1.482 | 11.18 | 0.24 | |
| | AC | 1.86 | 69.35 | 0.870 | 12.00 | 0.25 | |
| T13 | WR | 2.82 | 72.16 | 1.267 | 11.53 | 0.24 | |
| | | | | | | | 11.55±0.43 |
| *Granite* | | | | | | | |
| T11 | BC | 5.20 | 86.42 | 3.040 | 14.98 | 0.31 | |
| | AC | 2.12 | 80.36 | 1.316 | 15.90 | 0.35 | |
| | | | | | | | 15.46±0.42 |

[a] WR = whole rock; MC = mineral concentrate; BC = biotite concentrate; AC = amphibole concentrate.
[b] Percent of $^{40}Ar$ radiogenic.
[c] $cm^3$ STP/gr $\times 10^{-6}$ of radiogenic Ar.

## 2.3. Palaeomagnetism

### 2.3.1. Procedure

Our palaeomagnetic study on Tinos focused on ten sites in different dacitic dikes and three sites in the adjacent monzogranite pluton. The sites are scattered over the entire area of SE Tinos (Fig. 2A). Six field-drilled cores were collected from each site, oriented in the field by both sun compass and a Brunton compass. Remnant magnetization of all samples was measured using a three-axis 2-G super conducting magnetometer with integrated alternating field (AF) coils, at the Geophysical Institute of Israel, Holon (Israel). Samples were stepwise demagnetized by AF with peak field up to 120 mT, in 10-mT increments. Two samples from each site were stepwise thermally demagnetized up to 585°C in order to compare with the results of the AF treatment. Principal component analysis was done following Kirschvink's [24] method, and statistical data analysis was carried out following Fisher [25]. Dike dips were measured in the field, but because the majority of the dikes were within 5° from the vertical, no tilt correction was required. Since palaeohorizontal markers are absent from the monzogranite, no tilt correction was performed on these rock bodies too.

### 2.3.2. Results

The palaeomagnetic results are shown by orthogonal vector plots and normalized intensity curves ($J/J_0$) (Fig. 4). Some of the samples show two components of magnetization: a secondary component of moderate coercivity, typically removed by AF demagnetization at a peak field of 30–50 mT, or at a temperature of 300°C, and a stable characteristic component (Fig. 4, T2). The majority of the samples show a single magnetization component with high coercivity, removed only at temperatures of 530–585°C (Fig. 4, T9, T11). Samples that showed no stable vector throughout the demagnetization treatment, or directions that significantly deviated from the mean direction of the respective site, were rejected and excluded from the statistical calculations.

Site mean magnetization directions are plotted on a lower hemisphere equal area stereographic projection (Fig. 5) and shown in Table 2. We provide the mean of all the measured sites, as well as separate results from dikes and monzogranite. We believe that the secular variations were fully averaged, because the dikes and the monzogranite have different ages, and because both normal and reverse magnetizations were detected. Within-site dispersions are small ($\alpha_{95} < 10°$, $K > 60$, except site T13), but the dispersion of the site means about the mean directions are quite large ($K = 15.6$, $\alpha_{95} = 11.2°$ for

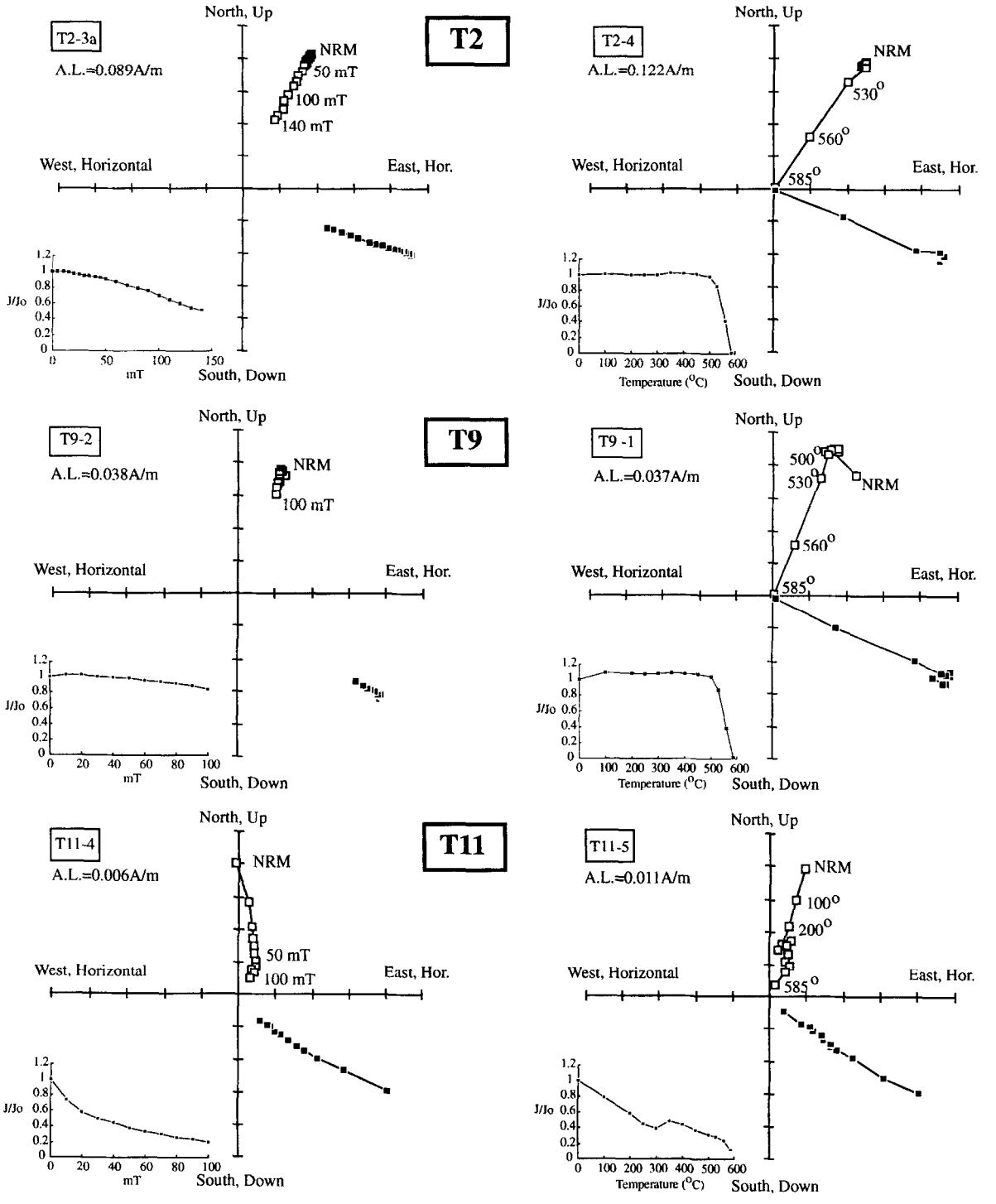



Fig. 4. Orthogonal vector plots and normalized intensity plots ($J/J_0$) of samples from three dikes. Left plots show samples that were stepwise demagnetized in an alternating field (AF); values are in milliteslas (mT). Right and bottom plots show thermally demagnetized samples; temperatures are in degrees Celsius. Open squares represent the declination component; solid squares represent the inclination component.

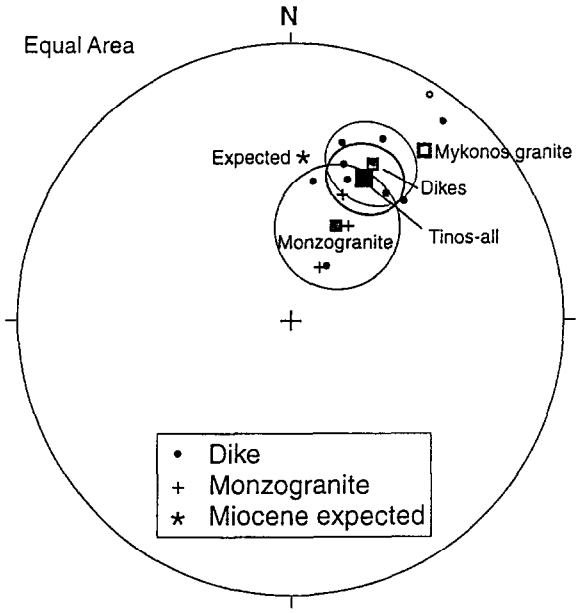



Fig. 5. Lower hemisphere, equal area stereographic projection showing the position of the mean remnant directions and associated $\alpha_{95}$ cones of confidence of Tinos dikes, tinos monzogranite, Tinos all sites (present study), and Mykonos granite (from Morris and Anderson [17]). Also shown is the position of the expected reference pole position ($D$ = 005°, $I$ = 42°) determined by Kissel and Laj [16] in Kimi area (Evia).

the dikes; $K$ = 46.1, $\alpha_{95}$ = 11.9° for the monzogranite) (Fig. 5). Remnant direction obtained on all dikes is: $D$ = 28.1°; $I$ = 36.7°. Three sites in the monzogranite yielded $D$ = 26°; $I$ = 58.9° (Fig. 6, Table 2). The mean palaeomagnetic vector obtained for all sites is $D$ = 27.8°; $I$ = 41.9°, and is almost identical to that of the dikes (Fig. 5). Thus, we use this *mean-of-all-sites* ($D$ = 27.8°; $I$ = 41.9°) in the following analysis of the rotational history of the island of Tinos.

### 2.3.3. Evaluating the deviation from the expected pole position

In order to evaluate the magnitude of vertical- and horizontal-axis rotation affecting Tinos, a comparison of our measured palaeomagnetic directions with an expected Miocene reference direction has been made. We first attempt to compare our results with the expected direction of Miocene pole position, on the basis of both African and European apparent polar wander paths (APW) of Westphal et al. [26]

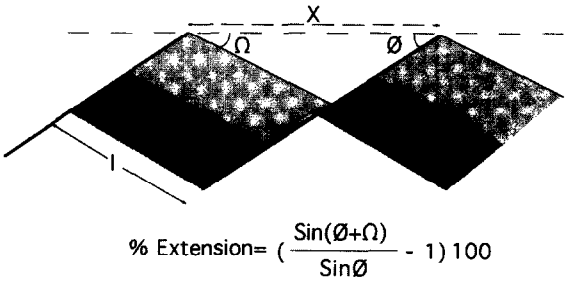



Fig. 6. Domino style geometry and derivation of the magnitude of extension on the basis of tilting ($\Omega$) and fault-plane orientation ($\Phi$) [33].

and Besse and Courtillot [27], which is $D$ = 005°, $I$ = 5°3.

Using Beck's [28] method with Demarest's [29] corrections, we obtain the following declination rotations ($R$) and inclination flattening ($F$): dikes only (10): $R$ = 23.1 ± 16.3° (clockwise), $F$ = 16.3 ± 12.3°; monzogranite (3): $R$ = 21.0 ± 24.9° (clockwise), $F$ = −5.9 ± 12.9°; all sites (13): $R$ = 22.8 ± 15.7° (clockwise), $F$ = 11.1 ± 11.1°.

The declination data provides strong indication for 23° clockwise rotation of the island of Tinos after the intrusion of the dikes at 11.5 Ma.

The inclination data is, however, more difficult to evaluate. The inclination flattening (11.1°) deduced from mean of all sites a priori suggests post-12 Ma northward drift of Tinos with respect to both Africa and Europe. However, this northward drift is at odds with current tectonic models that suggest southward motion of Aegea [2,10]. A similar inclination flattening has been previously observed in other palaeomagnetic studies of Tertiary rocks in the Aegean ([16,30] and references therein; [31]). Yet, such a flattening is difficult to reconcile with current knowledge on crustal deformation in the region because it implies a considerable amount (ca. 500–1000 km [16,30]) of post-Middle Miocene plate convergence at the northern boundary of the anomalous region, i.e. in the Rhodope area of northern Greece. However, rather than being a consuming plate margin, the Rhodope area has been subjected to crustal extension since the Oligocene [32] and a phase of post-Middle Miocene crustal thickening required to consume the drifted crust is not known. Thus, the overall nature of the inclination anomaly in the Aegean, including Tinos, remains obscure.

Table 2
Summary of palaeomagnetic results on Tinos

| Site | Lithology | $N/N_0$ | Inc. (°) | Dec. (°) | $\alpha_{95}$ (°) | $K$ |
|---|---|---|---|---|---|---|
| T1 | AD | 3/6 | −4.9 | 212.2 | 10.2 | 62.3 |
| T2 | AD | 6/6 | 27.5 | 27.2 | 3 | 361.4 |
| T3 | MG | 6/6 | 56.5 | 30.4 | 3.1 | 339 |
| T4 | AD | 2/5 | 10.2 | 37.9 | 7.1 | 195.3 |
| T5 | AD | 5/5 | 70.8 | 33.5 | 2.3 | 759.2 |
| T6 | AD | 5/5 | 44.1 | 22.3 | 8 | 61.2 |
| T7 | AD | 3/5 | 42.2 | 37.4 | 6.8 | 142.2 |
| T8 | AD | 3/5 | 40.1 | 43.5 | 8.4 | 92.3 |
| T9 | AD | 6/6 | 40.5 | 18.9 | 6.3 | 83.1 |
| T10 | AD | 6/6 | 47.9 | 9.3 | 7.3 | 62.2 |
| T11 | MG | 3/5 | 48.5 | 22.1 | 7.8 | 108.1 |
| T12 | MG | 5/5 | 71.6 | 26.7 | 5.4 | 133.6 |
| T13 | AD | 4/4 | 33.9 | 16.6 | 19.5 | 12.9 |
| Mean AD | AD | 10/10[a] | 36.7 | 28.1 | 11.2 | 15.6 |
| Mean MG | MG | 3/3[a] | 58.9 | 26 | 11.9 | 46.1 |
| Mean-all | | 13/13[a] | 41.9 | 27.8 | 9.9 | 15.3 |
| Expected | | 14[b] | 42 | 5 | 5 | 57 |

Lithology: AD = andesite dike; MG = monzogranite; $N$ = number of samples included in analysis; $N_0$ = number of measured samples; Inc. = inclination; Dec. = declination; $\alpha_{95}$ = radius of 95% confidence in degrees; $K$ = precision parameters.
[a] Number of sites.
[b] Number of studies used in the statistics.

For the purpose of structural interpretations the effect of this northward drift is corrected at present. We follow Morris and Anderson [17] and used a revised reference pole position of $I = 42°$ $D = 005°$ deduced from tilt-corrected sites by Kissel and Laj [16] in Kimi (Evia, northwestern Cyclades). The inclination flattening thus obtained on Tinos becomes practically zero.

### 2.4. Crustal extension by block rotation

#### 2.4.1. Tilting

The tectonic rotations which are implied by our measurements constrain the pattern of rigid-block deformation of the Cyclades in the area of Tinos island. The geochronological data (Table 1) indicate that these rotations took place after the intrusion of the dacitic dike swarm at 11.5 Ma, but the data do not permit to define the timing of this deformation more accurately.

Assuming that the dacitic dikes intruded as nearly vertical sheets, their present sub-vertical orientation implies that the post-12 Ma deformation of the area did not involve significant tilting. Tilting may be underestimated if the axis of rotation lay close to perpendicular to the dike walls, but we consider this option unlikely because that direction is perpendicular to the strike of the faults bounding Tinos and to the roughly NNE–SSW extension that prevailed in the Aegean at the time of the intrusion of the dikes and afterwards. In addition, the inclination flattening obtained by us using the reference pole position from Kimi (Evia; [16]) leaves little room for tilting.

Various types of deformation styles may lead to crustal extension, but one of the most typical modes of upper-crust continental extension involves normal faulting and tilting of blocks in a domino-style geometry (e.g. [33,34] and references therein). An important corollary of the planar geometry associated with the domino-style is that both, the faults and the blocks they bound, rotate about a horizontal axis. The absence of such tilting on Tinos implies that post-12 Ma crustal thinning in this area was minor.

#### 2.4.2. Vertical-axis rotation

Vertical-axis block rotation, commonly associated with wrench-faulting, is an important deformation

style of the continental crust (e.g. [35–37]). On mainland Greece, clockwise rotations were detected in several sites [16], whereas rotations on the Turkish side of the Aegean are less uniform, being both clockwise and anticlockwise. Kissel and Laj [16] suggested that the rotations in the Aegean region resulted in the development of the Hellenic arc and its southward migration above the Hellenic trench. Taymaz et al. [38] suggested that the westward motion of Turkey, the resistance to this motion in the northern part of Greece and in Albania, and the presence of the Hellenic subduction zone that acts as a free boundary, govern the overall southward flow of the Aegean crust. They depicted a series of NW–SE-trending crustal segments ('slats') in the western part of the Aegean, and NE–SW-trending slats in its eastern part. In this model, the deformation in the Aegean involved opposite sense of rotations of these crustal segments and was accompanied by roughly N–S extension.

Our palaeomagnetic study revealed 23° vertical-axis clockwise rotation of the 12-Ma-old dike swarm on Tinos. Post-10 Ma clockwise rotations were also detected on Mykonos [17] and in Evia [16]. These results support the general outlines of the model proposed by Taymaz et al. [38] and strengthen the possibility that Mykonos and Tinos lie on a NW–SE-striking crustal segment that stretches as far north as central Evia and rotated clockwise.

Vertical-axis block rotation is a two-dimensional crustal deformation mechanism that preserves area and crustal thickness, but modifies the length and width of the domain boundary [35,36]. McKenzie and Jackson [37] suggested a different mode of block rotation in which the length of the domain boundary is kept constant. In this case, the faults bounding each of the rotating blocks show oblique-slip with a component of extension perpendicular to the strike of the crustal blocks. The axis of rotation is thus inclined, and a tilting component is involved. Because no tilting was found on Tinos, the axis of clockwise rotation there is considered by us as a purely vertical one. Therefore, in what follows, we consider the mode of block rotation that preserves area, but modifies the length of the domain boundaries.

The procedure to calculate the N–S elongation due to vertical-axis block rotation resembles that applied in calculating extension due to a domino-style normal faulting (Fig. 6). The extension is given by:

$$\%\text{Extension} = ((\sin(\Phi + \Omega)/\sin\Phi) - 1)100$$

where $\Phi$ is the angle between the fault plane and the N–S direction, and $\Omega$ is the rotation angle.

In calculating the N–S elongation of the Cyclades, we consider that the western Cyclades composed of a set of crustal slats bounding by NW-striking faults of the type proposed by Taymaz et al. [38]. The angle $\Phi$ between the NW-striking faults and the N–S direction is ca. 45°. As indicated by our palaeomagnetic data on Tinos, the angle of rotation $\Omega$ is 23°. The N–S elongation thus obtained is 31%.

The N–S dimension of the Cyclades (i.e. the distance between Tinos and Santorini; Fig. 1) is at present ~100 km long. Our calculation implies that about 25 km of this length are due to post-12-Ma vertical-axis block rotation. No major crustal thinning is required to have accompanied this phase of N–S elongation in the western Cyclades.

## 3. Post-Middle Miocene extension due to low-angle normal faulting in the central Aegean

Post-Middle Miocene low-angle normal faults are observed in a restricted extensional corridor encompassing Mykonos, Paros and Naxos in the central part of the Cyclades (Fig. 7). They usually cut through Middle Miocene (10–12 Ma; [5]) granitoids in the footwall, and place various types of sediments and volcanics in the hangingwall. Sedimentary sequences, though allochthonous, are preferentially preserved in association with deformed granites and are absent from the rest of the western Cyclades.

Although the low-angle extensional structures (including the low-angle detachment on Tinos) are important contributors to the Late Cenozoic continental deformation of the Cyclades, there is little structural evidence concerning the original dip of these structures across which a crustal section, sometimes tens of kilometers thick, has been cut out [4]. In the Basin and Range province, controversy exists over whether detachments accommodate large-magnitude extensional deformation. One set of models suggests that they were originally high-angle normal faults that were tilted to sub-horizontal attitudes [39]. Alternatively, at least some of these detachments may

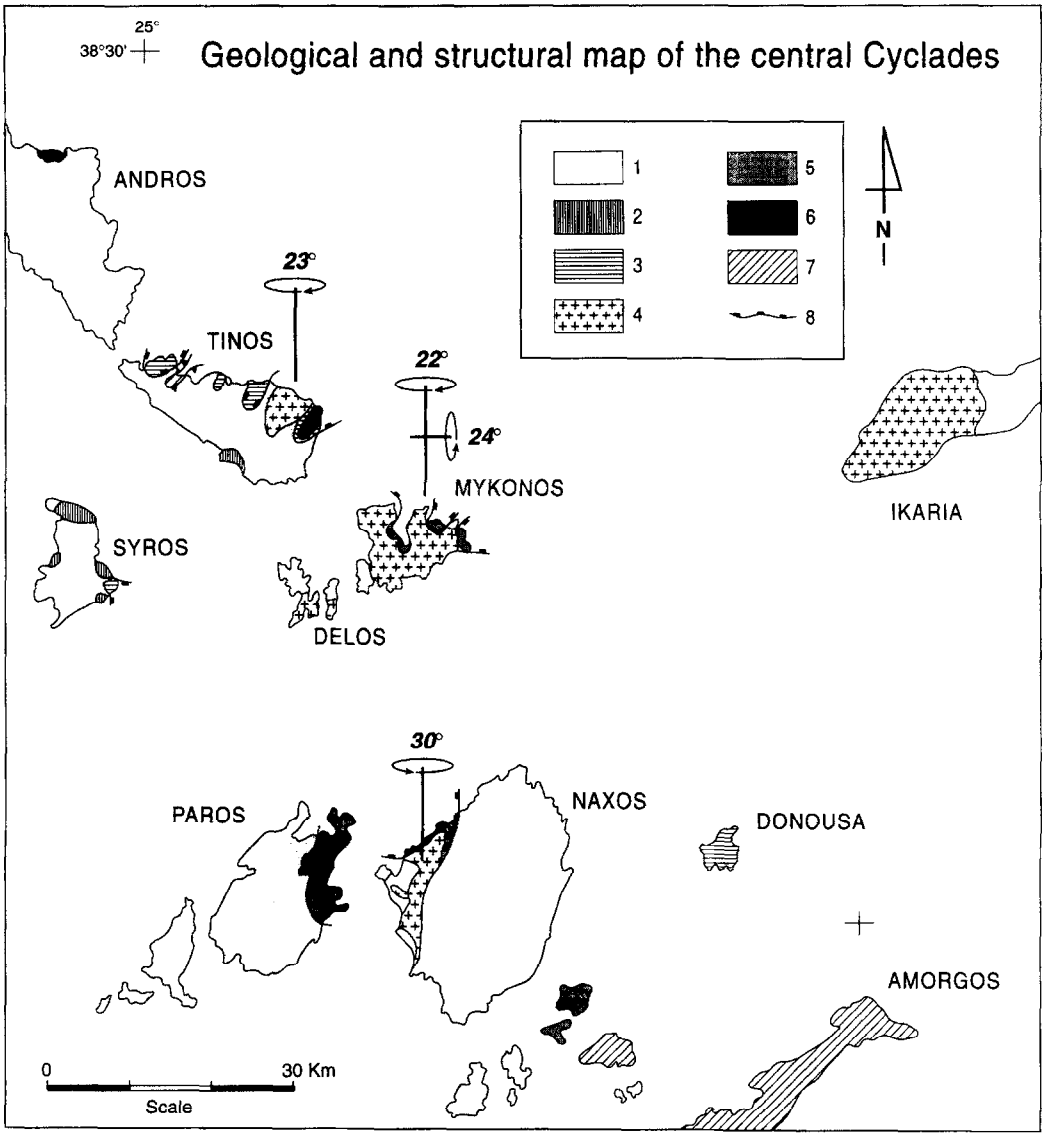



Fig. 7. Structural map of the central Aegean region (area marked on Fig. 1) showing the distribution of low-angle normal faults [4], and palaeomagnetic rotations on Tinos (present study), Mykonos and Naxos [17]. Post–Middle Miocene low-angle normal faults concentrate in the central part of the Cyclades, in the area encompassing Mykonos, Paros and Naxos. These low-angle normal faults cut through Middle Miocene granitoids in the footwall and place various types of sediments in the hangingwalls. Pre–Middle Miocene low-angle fault is pierced by a granite on Tinos island. Key: *1* = cycladic blueschist unit reworked in greenschist and amphibolite facies; *2* = well-reserved eclogite facies rocks; *3* = L–P metamorphic Upper Unit; *4* = Middle Miocene granitoids; *5* = Early Miocene clastic sediments of the Upper Unit; *6* = ophiolites of the Upper Unit; *7* = basal unit; *8* = low-angle normal fault. Arrows mark sense of ductile shear (data sources in the text).

have originated as low-angle structures (for example [40]). In what follows, we use structural and published palaeomagnetic data to deduce the origin of the low-angle normal fault exposed on Mykonos island.

### *3.1. Low-angle normal faulting on Mykonos island*

Mykonos island (Fig. 8) is made of an I-type granite [8], small patches of the metamorphic country rock, and remnants of an allochthonous upper

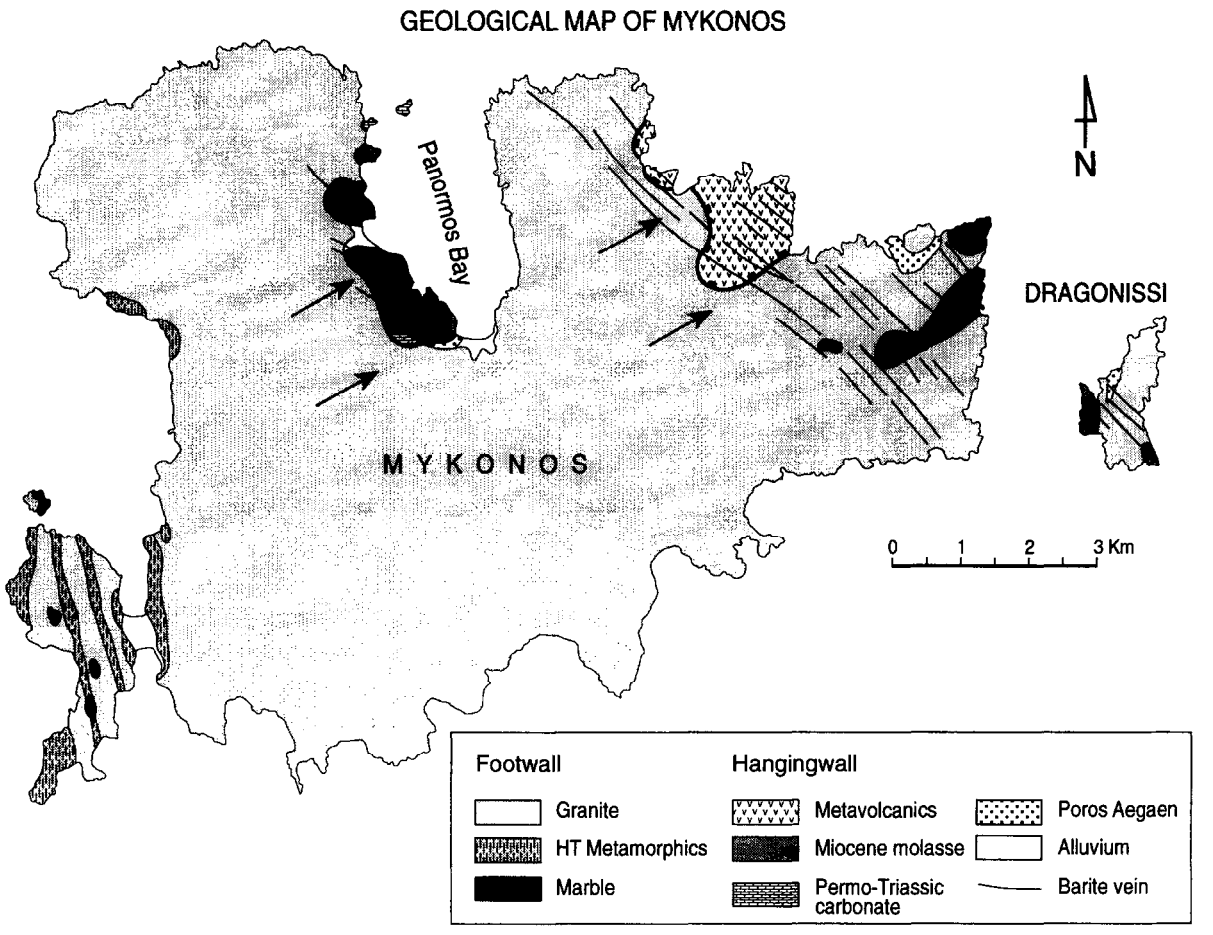



Fig. 8. Geological map of the island of Mykonos (slightly modified after Vorgeadis [44]). The island consists of a 10–12-Ma granite [8] intruded into a metamorphic country rock (particularly exposed on the SW border of the pluton). The granite is tectonically overlain by a sequence of coarse-grained detrital sediments with small slices of Permo–Triassic carbonate. The age of detrital sedimentation is Oligo-Miocene [43]. The contact between the granite and the overlying sediments is a low-angle normal fault. The granite shows increasing intensity of deformation towards the contact and its mylonitic foliation has been overprinted by cataclasis near and at the contact. Sense of shear criteria and lineations (arrows) indicate top-to-the-ENE motion [41,42]. After extensional motion ceased, the low-angle normal fault has been cut by a vertical set of barite veins.

unit containing molasse sediments and greenschist-facies metavolcanics. K–Ar and Rb–Sr data on magmatic minerals (hornblende and biotite) in the granite yielded ages of 10–12 Ma, and apatite fission track gave ages of 9.5–10 Ma [8] reflecting rapid cooling during extension [41,42].

A low-angle normal fault inclined 30° towards NE is exposed on Mykonos (Fig. 8). Shallow-dipping foliation associated with a NE-trending stretching lineation defined by ribbon quartz and partly recrystallized feldspar augens developed in large parts of the footwall granite and top-to-the-ENE sense of shear criteria predominate [41,42]. The ductile shearing fabric is overprinted by cataclasis near the contact where the granite is strongly deformed and brecciated. In the area of Panormos Bay, the hangingwall consists of a series of molasse-type clastic sediments whose age has been estimated as Oligo-Miocene [43]. Sedimentary layering in the hangingwall dips towards the fault plane, ca. 30° to the SW (Fig. 9).

Both, the granite and the overlying sediments, were pierced by a conspicuous set of NW-trending tension gashes [44] filled with barite (see Figs. 8 and

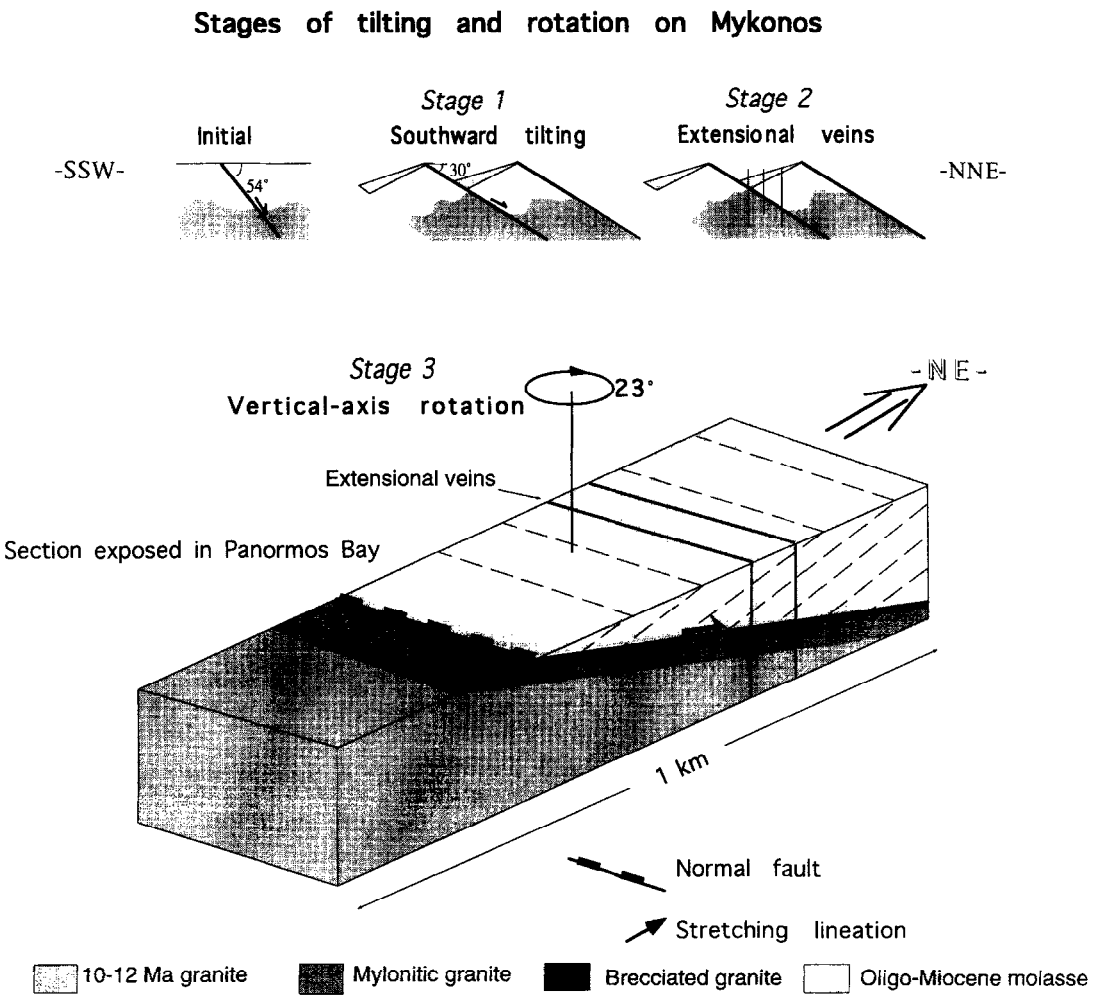



Fig. 9. Block diagram demonstrating the structure of the low-angle normal fault on Mykonos (in the area of Panormos bay). Domino-style normal faulting involving 24° rotation over a horizontal axis affected hangingwall sediments, footwall monzogranite and the intervening normal fault. A set of barite veins postdated tilting and extensional motion on the fault, but shared the subsequent clockwise rotation of the entire island.

9). These barite veins are vertical, usually several meters thick, and can be traced over hundreds of meters along strike (Fig. 8). They are slightly oblique to the stretching lineation in the granite. Since these veins are nowhere displaced by the low-angle normal fault, they postdate the extensional motion on this fault. Nevertheless, their NW–SE trend suggest that they were involved in the vertical-axis rotation that affected Mykonos (see below). Thus, they prove an excellent and unique marker to resolve the time relations between tilting and vertical-axis rotation inferred from the palaeomagnetic data.

*3.1.1. Constraints from palaeomagnetic results from Mykonos*

In a recent palaeomagnetic study of the footwall granite, Morris and Anderson [17] found significant deviations of both inclination and declination from the expected Miocene direction. They obtained inclination values, $I = 23°$, and declination value, $D = 38°$. Morris and Anderson compared their results to the expected vector obtained by Kissel and Laj [16] on the basis of tilt-corrected data in Kimi. The use of this pole position from Kimi ($D = 005°$, $I = 42°$) minimizes the effect of the (apparent?)

northward drift and the entire inclination shallowing thus obtained is due to tilting. Nevertheless, Morris and Anderson [17] faced difficulties in resolving the orientation of the rotational axes because a palaeohorizontal marker was not defined. They examined several possible rotational paths, each yielding different magnitudes of tilting and rotation depending on the order of which these rotational components are applied. As a whole, Morris and Anderson calculated southward tilting of either 27 or 37° and clockwise rotation of either 13 and 22°, and argue that 22° is the most likely magnitude of clockwise rotation that affected Mykonos.

The structural relations of the barite veins on Mykonos have been used by us to better resolve the rotational history of this island. A key observation is that the dacitic dikes on Tinos and the barite veins on Mykonos have parallel orientations, both are vertical and strike NW–SE. These two extensional markers are prominent on the island scale and therefore are likely to have been formed in response to a regional stress field. The fact that these two markers are still parallel implies that they shared an identical amount of vertical-axis clockwise rotation, as, indeed, is confirmed by the palaeomagnetic data on both islands. Similarly, the identical upright orientation of the dikes and the barite veins indicates that they were not affected by tilting. The fact that the barite veins remain vertical, in spite of the important amount of tilting that has been detected by the palaeomagnetic data on the granite in Mykonos [17], implies that these veins penetrated into an already-tilted structure. They escaped tilting, and have subsequently shared only the clockwise vertical-axis rotation with their host rock. Thus, on Mykonos, tilting and motion on the low-angle normal fault were separated from the subsequent vertical-axis block rotation by a phase of extensional veining. This separation allows us to treat each of the rotational components (horizontal and vertical) independently, in the correct order of events.

Therefore, a counterclockwise rotation by 23° should first be applied to the palaeomagnetic direction obtained by Morris and Anderson ($I = 23°$, $D = 38°$) in order to resolve the amount of tilting on Mykonos (Fig. 10). Then, the remnant palaeomagnetic direction is restored to the reference direction ($I = 42°$, $D = 005°$) by re-tilting of ca. 24° about a horizontal axis trending 330°. In the restored position the (now) low-angle fault on Mykonos is much steeper, dipping 54° towards 025°, and the mylonitic lineation is oriented 50°/027°.

This procedure also retilts the sedimentary layering to sub-horizontal position, as should be expected. The above results suggest that the (now) low-angle normal fault on Mykonos originated steeper, at a dip of ca. 54°. This is well within the range of dips of seismically active normal faults in the Aegean and in other extending terranes [34].

*3.1.2. Amount of extension on Mykonos*

Tilting during extension may be due to rotation on a set of planar normal faults or due to normal faults having a listric geometry. From the geometry of the structures on Mykonos, it is impossible to distinguish which of these mechanisms was dominant. Since the fault plane exposed is planar and as the broad structural features on Mykonos (especially the dual rotation of both hangingwall and footwall) are a priori consistent with a planar rotational normal faulting, we use this 'domino-style' of geometry in order to deduce the features of the extension on Mykonos.

The relatively simple geometry of planar normal faults allows to calculate the extensional strain if the angle of tilt and the dip of the fault-plane are known (Fig. 6). On Mykonos, tilting of 24° was estimated from the palaeomagnetic data, and the dip of the fault-plane is 30°. The resulting extension thus calculated is ca. 60%.

The magnitude of the dip-slip motion, as well as the width of the rotating block can also be estimated. The low-angle normal fault on Mykonos juxtaposes near-surface sediments on top of a ductile-deformed granite. Judging from the wide-regional abundance of Middle Miocene granitoid intrusions in the area, it is safe to assume that heat flow in the central Cyclades was then relatively high, and that the transition to ductile deformation regime took place at depths no greater than 10 km. Assuming an original fault-plane dipping 54°, a dip-slip motion of 12 km is implied by the juxtaposition of the 10-km depth crustal level against the sediments. If ductile conditions were attained at shallower crustal levels, dip-slip motion was even smaller.

Once dip-slip motion and the angular relations

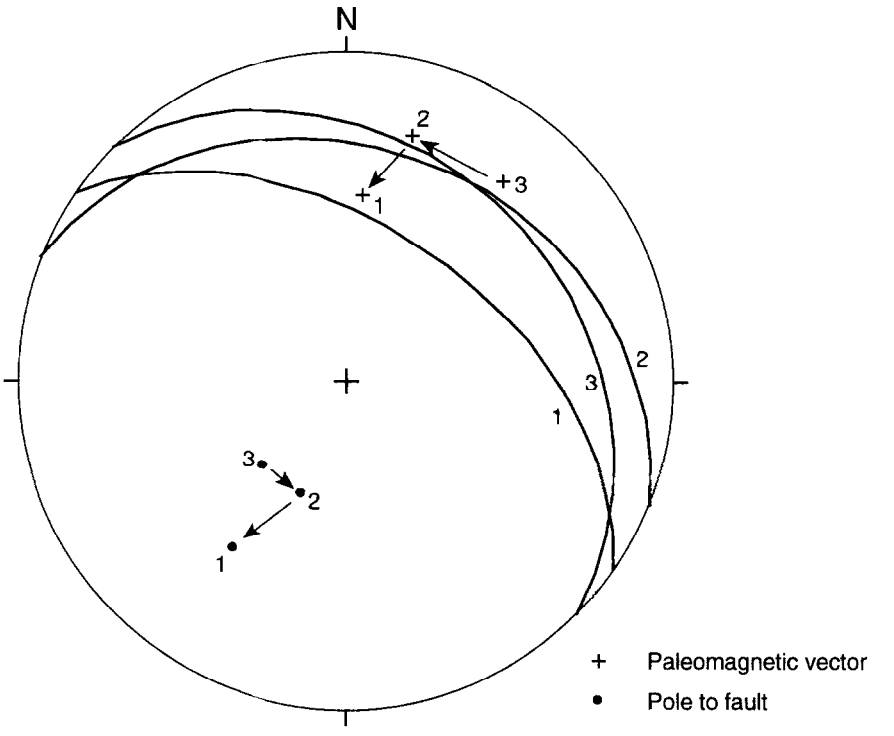



Fig. 10. Equal area stereographic projection showing the restoration of the low-angle fault and the mean palaeomagnetic vector obtained by Morris and Anderson [17] in Mykonos, using Kimi pole position as a reference. *3* = observed palaeomagnetic vector and present orientation of fault; *2* = observed palaeomagnetic vector and fault after 23° of counterclockwise rotation about a vertical axis; *1* = orientation of expected palaeomagnetic vector, and inferred original orientation of the fault.

are determined, a minimum block width can be calculated. On Mykonos, estimated block width thus obtained is 10 km, similar to the amount of dip-slip motion (Fig. 11). This implies that the southern half of the island of Mykonos (south of Panormos bay) constituted a southward-tilted upper crustal sliver.

The configuration presented in Fig. 11B shows that at the cessation of tilting and extension the part of the low-angle normal fault now exposed on Mykonos comprised the depocenter of a ca. 5-km-deep half graben. If this half graben was filled with sediments in the Middle Miocene, the present exposure implies the removal of a thick rock column and significant uplifting. Uplifting is also inferred if the present level of exposure was kept near sea level throughout the extensional history. One possible mechanism to reduce the depth of the half-graben basin is by inducing rotation and uplifting of the normal fault in response to buoyancy forces accompanying denudation of the footwall [45,46].

## 4. Summary and conclusions

Our study was aimed to constrain the mode of post-12 Ma crustal extension in the Cyclades area, particularly in the vicinity of the islands of Tinos and Mykonos. On Tinos, we mapped a swarm of subvertical NW-trending dacitic dikes that penetrated Alpine metamorphic rocks. K–Ar analyses yielded ages of $11.5 \pm 0.4$ Ma indicating that these dikes witnessed a substantial part of the extensional history of the central Aegean region. The palaeomagnetism of the dacitic dikes and the adjacent 15–18 Ma granitoid show 23° vertical-axis clockwise rotation and inclination flattening of ca. $11 \pm 11°$. We assume that the dacitic dikes intruded near-vertical (as they are today) and that no tilt correction is required. We thus suggest that the inclination flattening pertains to a regional-scale inclination anomaly which points to (an apparent?) northward drift of crustal blocks in the Aegean [30]. Using, instead, a tilt-corrected expected direction from Evia ($I = 42°$ $D = 005°$;

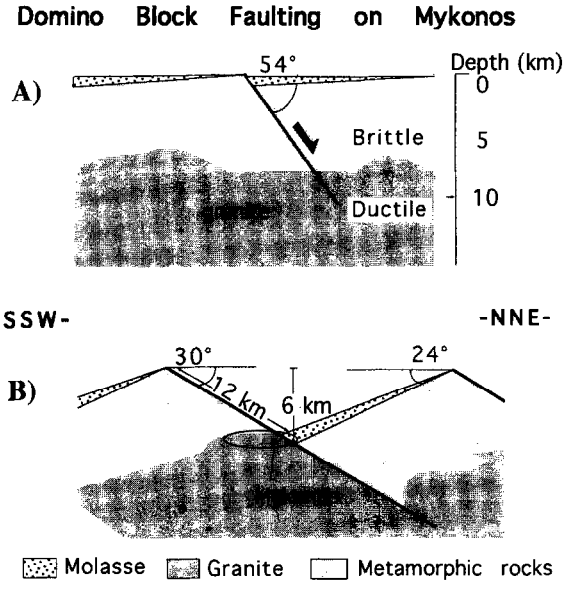



Fig. 11. Geometrical determination of the magnitude of dip-slip motion, width of tilted blocks, and the shape of the half-graben assuming a simple model of 'domino-style' extension. (A) Initial fault plane dipping 54°. The ductile deformation in the exposed granite footwall suggests a vertical separation of ~10 km (depending on the depth to the brittle–ductile transition). (B) After 24° of southward tilting. Dip-slip motion of 12 km is estimated from geometrical relations. We note that the present structure of Mykonos is more complex than illustrated here due to intensive late (steep) faulting and fracturing. In addition, there is little data on the real shape and size of the granite at depth, but the widespread Miocene magmatism suggests that these rocks invaded significant parts of the middle and upper crust.

[16]), the amount of flattening is reduced to zero. Assuming planar-rotational fault geometry, the absence of tilting implies that the amount of post-12 Ma extension due to normal faulting in the area of Tinos island was minor.

On Mykonos, a low-angle normal fault dipping 30° to the NE cuts through a 10–12-Ma granite in the footwall. The hangingwall is composed of an Oligo-Miocene molasse series whose layering dips 30° to the SW. Recent palaeomagnetic results [17] revealed 22° of clockwise rotation of the footwall granite and a considerable amount of tilting (when compared with Kimi pole position). A set of vertical NW–SE-striking barite veins (tension gashes) that penetrated the low-angle normal fault was used by us to show that on Mykonos, tilting predated vertical-axis rotation, and that the two rotational components were separated by a phase of extensional veining. It was also demonstrated that normal-faulting was accompanied by 24° of southward tilting, and hence the (now) low-angle (ca. 30°) normal fault on Mykonos originated dipping 54° towards 025°.

Restoration of the L–S microstructures indicates that the ductile fabric in the footwall granite originated dipping ca. 40–50°. This suggests that the normal fault on Mykonos merged with a moderately dipping ductile shear-zone at depth.

Using planar normal fault geometry, crustal extension of ca. 60% has been calculated on Mykonos, with dip-slip motion of about 12 km, and block width of a similar dimension. The magnitude of the dip-slip motion and the exposure of ductile deformed rocks suggest that Mykonos is a tilted upper-crustal sliver.

Previous workers suggested that the allochthonous sediments on Mykonos represent relics of a far-traveled extensional nappe originated in the southern Aegean [41,47]. The above data indicate, instead, that only a modest amount of displacement (ca. 12 km) is required on the (now) low-angle normal fault, and that the Oligo-Miocene sedimentary sequences were derived from a nearby site as previously suggested by Gautier and Brun [5].

Whereas 60% extension is inferred on the island of Mykonos, the absence of tilting on the neighboring island of Tinos implies that the amount of post-12 Ma extension there was minor. In spite of these significant differences, crustal thickness below the entire area changes only slightly, from ca. 30 km below Tinos to possibly 28 km below Mykonos [1]. This suggests that upper-crustal levels were decoupled from the lower crust. Tilting and extension in the upper crust may have been separated from a rigid lower crust by a low-angle detachment. However, the widespread S- and I-type granitic magmatism in the Cyclades [8] indicate that high temperatures and melting prevailed in the lower crust in the Middle Miocene, and therefore, that the deep crust was then mobile and rheologically weak. Thus, it is more likely that the uniform crustal thickness and the smooth Moho in the area were maintained by lateral flow of lower or middle crust material. Flow of mid crustal material from below rigid blocks to replace extended upper crust has been previously suggested to explain the smooth Moho observed in the Basin and Range province [48,49].

Both the dacitic dikes and the barite veins sealed older low-angle extensional structures and mark the cessation of a major extensional phase at around 10 Ma. Subsequent crustal deformation that dominated each of these islands and probably the entire Cyclades was governed by vertical-axis block rotation. A recent study [50] showed that Miocene strike–slip faults, that may have accommodate block rotation, are much more common in the Cyclades than hitherto thought. We estimated that post-10-Ma vertical-axis rotation increased the N–S dimension of the Cyclades by ca. 25 km. Block rotation affected the Cyclades when crustal thickness was restored to normal and its gravitational potential relatively reduced. Thus, vertical-axis rotations were probably induced by the westward extrusion of Turkey and/or the southward retreat of the subduction zone south of Crete, both acting on the periphery of the Aegean domain.

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